Empirical Strategy

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1 Identification Problem

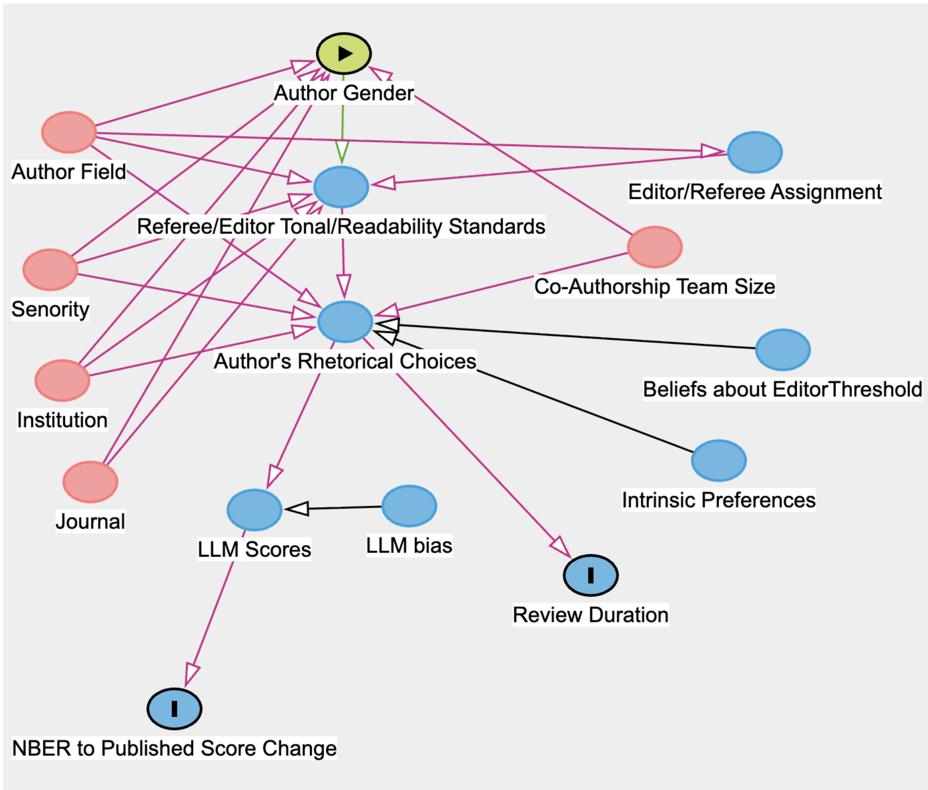
This project asks whether the peer-review-driven divergence documented in Hengel (2022) for readability appears across a broader set of rhetorical and tonal dimensions. The central problem in this project is therefore identical to that of Hengel (2022): we observe only published versions' abstracts and, for a certain subset, NBER draft versions. Any measured gap in readability or tone could reflect higher standards in publishing for women, but it also could reflect:

- Pre-existing gender differences in writing style driven by risk aversion or preference.
- Differences in topic, field, or research that correlate with both author gender and writing style
- Selection in Publication: female-authored papers that were published may be systematically different, even in the absence of a higher standard
- Bias in the LLM evaluations

The key to the empirical analysis in this project is comparing the NBER working paper to the published article. If the differences in tone and readability are larger in the published versions than the working paper versions, then this is evidence of different standards being applied in peer review to men and women.

2 DAG

Figure 2.1: Directed Acyclic Graph



3 Key Model Equations

All equations parallel the **?** specification, substituting LLM evaluations and group composite scores as outcomes in place of (or alongside) readability scores. The treatment variable throughout is *Female Authorship*, defined as:

$$FA_j = \begin{cases} 0 & \text{if } \dfrac{\text{female authors}_j}{\text{total authors}_j} < 0.50 \\ \dfrac{\text{female authors}_j}{\text{total authors}_j} & \text{if } \dfrac{\text{female authors}_j}{\text{total authors}_j} \geq 0.50 \end{cases} \tag{1}$$

Papers with fewer than 50 percent female authors are classified as male-authored ($FA_j = 0$); for majority female-authored papers, FA_j is the continuous female author share. This threshold follows Hengel (2022), who documents that the readability gap between male- and female-authored papers begins to emerge at this proportion.

3.1 LLM Group Composites

The 16 rubric criteria are grouped into five composite scores, each scored on a 1–10 scale:

$$G1_j = \frac{\texttt{modal\_verb}_j + \texttt{hedging}_j + \texttt{qualifier}_j + \texttt{ack\_limits}_j + \texttt{caution}_j}{5} \qquad \text{(Creativity \& Hedging)}$$

$$G2_j = \frac{\texttt{assertiveness}_j + \texttt{active\_passive}_j}{2} \qquad \text{(Assertiveness \& Voice)}$$

$$G3_j = \frac{\texttt{directness}_j + \texttt{imperative}_j}{2} \qquad \text{(Structural Directness)}$$

$$G4_j = \frac{\texttt{pronoun}_j + \texttt{novelty}_j + \widetilde{\texttt{jargon}}_j + \texttt{emotional}_j}{4} \qquad \text{(Authorial Stance \& Novelty)}$$

$$G5_j = \frac{\texttt{evidence}_j + \texttt{practical}_j}{2} \qquad \text{(Support \& Impact)}$$

where $\widetilde{\texttt{jargon}}_j = 11 - \texttt{jargon}_j$ is a scale-flipped version of the raw jargon score, so that higher values correspond to lower technical density, consistent with the direction of all other criteria. Higher values on all five composites reflect a stronger presence of the named stylistic attribute.

3.2 Article-Level OLS

The first estimating equation regresses each published LLM composite score on the female authorship share of the paper, conditional on a large set of controls:

$$G_{jP}^{(n)} = \beta_0 + \beta_1\, FA_j + \mathbf{\Theta}'\mathbf{X}_j + \varepsilon_j \tag{2}$$

for $n = 1, \ldots, 5$, where $G_{jP}^{(n)}$ is the group n composite score for the *published* version of article j. The coefficient of interest, β_1, captures the average difference in published tonal profile between male- and female-authored papers, conditional on controls. The vector $\mathbf{X}_j$ includes editor fixed effects, a blind-review indicator, journal $\times$ year fixed effects, number of co-authors (N_j), dynamic institution fixed effects, citation count (asinh-transformed), author prominence ($\max T$), author seniority ($\max t$), a native-speaker indicator, and primary and tertiary *JEL* category dummies. Standard errors are clustered at the editor level.

Equation (2) is estimated separately for each of the five group composites and for each of the 16 individual criteria. This is a cross-sectional estimate on the published sample and does not isolate the contribution of peer review to any gap.

3.3 First-Differenced OLS

To identify the portion of the tonal gap that forms *during* peer review, I match published articles to their NBER working paper drafts and estimate the change in the composite score on the female authorship share:

$$G_{jP}^{(n)} - G_{jW}^{(n)} = \beta_{0P} + \beta_{1P}\, FA_j + \mathbf{\Theta}_P'\mathbf{X}_{jP} + \mu_{jP} + \varepsilon_{jP} \tag{3}$$

where $G_{jW}^{(n)}$ is group n's composite score for the NBER working paper version of paper j, and μ_{jP} is an unobservable component assumed to have zero partial correlation with FA_j. Subtracting the working paper score from both sides of the published version regression eliminates the correlation between draft tonal style and female authorship that would otherwise bias OLS estimates of β_{1P}. The coefficient β_{1P} captures the additional tonal change during peer review for female-authored papers relative to male-authored papers. The NBER-matched sample covers approximately 1,988 working paper–article pairs.

3.4 Feasible GLS System

An alternative strategy based on Ashenfelter and Krueger (1994) jointly estimates the gender differences in the draft and final versions using generalised least squares, identifying the contemporaneous effect of peer review post-estimation by subtracting coefficients. The working paper equation is:

$$G_{jW}^{(n)} = \beta_{0W} + \beta_{1W}\, FA_j + \mathbf{\Theta}_W'\mathbf{X}_{jW} + \mu_{jW} + \varepsilon_{jW} \tag{4}$$

where β_{1W} reflects female authorship's impact on tonal style before peer review. Substituting a general structure for the correlation between μ_{jW} and observable controls into equation (4) and solving for the reduced form yields the published article equation:

$$G_{jP}^{(n)} = (\beta_{0W} + \beta_{0P}) + (\beta_{1W} + \beta_{1P})\, FA_j + \widetilde{\mathbf{\Theta}}_W'\mathbf{X}_{jW} + \widetilde{\mathbf{\Theta}}_P'\mathbf{X}_{jP} + \mu_{jP} + \widetilde{\varepsilon}_{jP} \tag{5}$$

Equations (4) and (5) are estimated jointly via feasible GLS. The peer-review-driven gap β_{1P} is recovered by subtracting the estimated coefficient on FA_j in (4) from that in (5).

3.5 Revision Duration

A separate strategy Hengel (2022) uses to evaluate whether women are held to a higher standard is by evaluating the duration of the peer review. She does this by using readability as a proxy for paper quality and controlling for it. I replicate this test in 2 ways:

Duration Test I: LLM Readability as Quality Control. The primary specification replicates Hengel (2022) using the LLM readability:

$$\begin{aligned} \mathit{Duration}_j = {} & \beta_0 + \beta_1\, \mathit{FA}_j + \beta_2\, \mathit{Pages}_j + \beta_3\, N_j + \beta_4\, \mathit{Order}_j \\ & + \beta_5\, \mathit{Citations}_j + \beta_6\, \mathit{Readability}_j + \beta_7\, \mathit{Theory}_j + \beta_8\, \mathit{Empirical}_j \\ & + \beta_9\, \mathit{Other}_j + \mathbf{\Theta}'\mathbf{X}_j + \varepsilon_j \end{aligned} \tag{6}$$

where $\mathit{Duration}_j$ is the number of days (or months) between first submission and final acceptance, $\mathit{Readability}_j$ is the LLM readability score for article j, and all other variables are as defined above. The vector $\mathbf{X}_j$ includes editor, year, and institution fixed effects. Standard errors are clustered by submission year. If women face higher standards, $\beta_1 > 0$: female-authored papers spend longer under review even after conditioning on writing clarity.

Duration Test II: Tonal Profile as Predictor. The second specification asks whether the tonal profile of a paper independently predicts review duration, and whether this effect differs by gender. For each group $n = 1, \ldots, 5$:

$$\begin{aligned} \mathit{Duration}_j = \beta_0 &+ \beta_1\, \mathit{FA}_j + \beta_2\, \mathit{Pages}_j + \beta_3\, N_j + \beta_4\, \mathit{Order}_j \\ &+ \beta_5\, \mathit{Citations}_j + \beta_6\, \mathit{Readability}_j + \beta_7\, G_j^{(n)} + \beta_8 \left(\mathit{FA}_j \times G_j^{(n)} \right) \\ &+ \beta_9\, \mathit{Theory}_j + \beta_{10}\, \mathit{Empirical}_j + \beta_{11}\, \mathit{Other}_j + \mathbf{\Theta}'\mathbf{X}_j + \varepsilon_j \end{aligned} \tag{7}$$

where $G_j^{(n)}$ enters alongside $\mathit{Readability}_j$. The coefficient β_7 captures whether the tonal dimension n predicts review duration on average; β_8 captures whether this effect differs by gender.

4 Core Assumptions

1. **Female Authorship Threshold**: Equation 1 is a modeling choice from Hengel (2022) and not directly testable.

2. **Conditional Mean Independence**: There is zero partial correlation between Female Authorship and the unobserved preferences conditional on the controls.

3. **NBER Draft Timing**: Since working papers are used as pre-submission drafts, they must actually reflect the paper and abstract at the time before peer review

4. **LLM Scoring Unbiasedness Across Gender**: Since I gave the LLM no information on the gender of the author, there is no way that the evaluations can be gender biased.

5. **LLM Score Stability**: Since we are comparing NBER abstracts to the published versions, the LLM must apply the rubric consistently across abstracts.

5 Threats to Identifications

5.1 LLM Evaluations

A major threat to identification in this project is the possible bias introduced by the LLM evaluations. To address this, I have engineered the prompt to avoid gender leakage and remain stable across different runs. I also have robustness checks planned to run the evaluations with different models and compare coefficients, which, if similar, indicate that LLM bias is not affecting our results.

5.2 Omitted Variable Bias

Since there is no canonical identification strategy, any causal interpretation must assume that the key equations control for any biasing variables. It is difficult to argue this fully, but I address this by controlling for the most obvious variables and by being careful with claiming a fully causal relationship.

5.3 NBER Response Times

As mentioned before, if the NBER working paper is posted after the author submits their paper for review, then the NBER paper reflects some peer-review feedback, masking the true effect of the review. Hengel (2022) empirically validated this assumption for the articles in her sample.

6 Planned Robustness Checks

1. **Alternative Definitions of Gender**: Following Hengel (2022), replicate all primary regressions using:(a) Solo-authored papers only, (b) Binary: At least one female author vs. none, (c) Binary: Over half female vs. none, (d) Single-gender papers only, (e) Senior female co-author vs. entirely male-authored papers

2. **JEL Fixed Effects**: Estimate key equations with (i) no JEL controls, (ii) primary JEL fixed effects, (iii)theory/empirical/other dummies, and (iv) tertiary JEL fixed effects.

3. **Individual LLM Criteria vs. Group Composites**: In addition to the five group composites, estimate key equations for each of the 16 individual criteria separately

4. **Multi-Model LLM Evaluation**: Re-evaluate a subsample of abstracts using an alternative model (e.g., GPT-4o or Gemini) and compare the distribution of gender gaps in LLM scores across models.

References

Ashenfelter, O. and Krueger, A. B. (1994). Estimates of the economic return to schooling from a new sample of twins. *American Economic Review*, 84(5):1157–1173.

Hengel, E. (2022). Publishing while female: Are women held to higher standards? evidence from peer review. *The Economic Journal*, 132(648):2951–2991.